



Islamic University Of Technology

EEE 4308 : Digital Electronics

Project Name : SAP – 1

Team : Digital Electronauts

Phase - C

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Team Members:

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Objective:

- 1) Learning about Registers of SAP-1
- 2) Learning about Memory unit of SAP-1
- 3) Gaining knowledge about D-latch
- 4) Gaining knowledge about D Flip-flop
- 5) Making sequential logic circuit

Description:

This project is about understanding of 6 bit computer and construction of it. Basically its about understanding inside process of a computer. We can understand what does it mean clock speed of a computer and how we can control this speed. For running this computer we need to program the RAM first, then we can run the computer. Program is stored inside RAM as a set of instruction. Other modules acts according to instruction. Program counter counts the instruction and MAR sends the memory address for next instruction. Instructed operation is occurred inside ALU. Value of ALU is stored inside Accumulator. When necessary , Accumulator sends the data to output register to show the result. Necessary signal for all the module is sent by controller sequencer. Controller sequencer generates signals according to instruction code sent by instruction register. All the module is runed by clock pulse. Without clock pulse this computer can't work. There is HLT operation in our computer that stopes the main clock, as a result whole computer stopes. This is total working system of our SAP-1.

Phase - C

Control Unit:

1) Instruction Register:

It is a 6-bit register which loads instruction from RAM through bus and splits it into two parts upper nibble and lower nibble. Then it sends upper nibble which contains opcode to controller sequencer for generating necessary signals and it sends lower nibble to bus which is received by MAR.

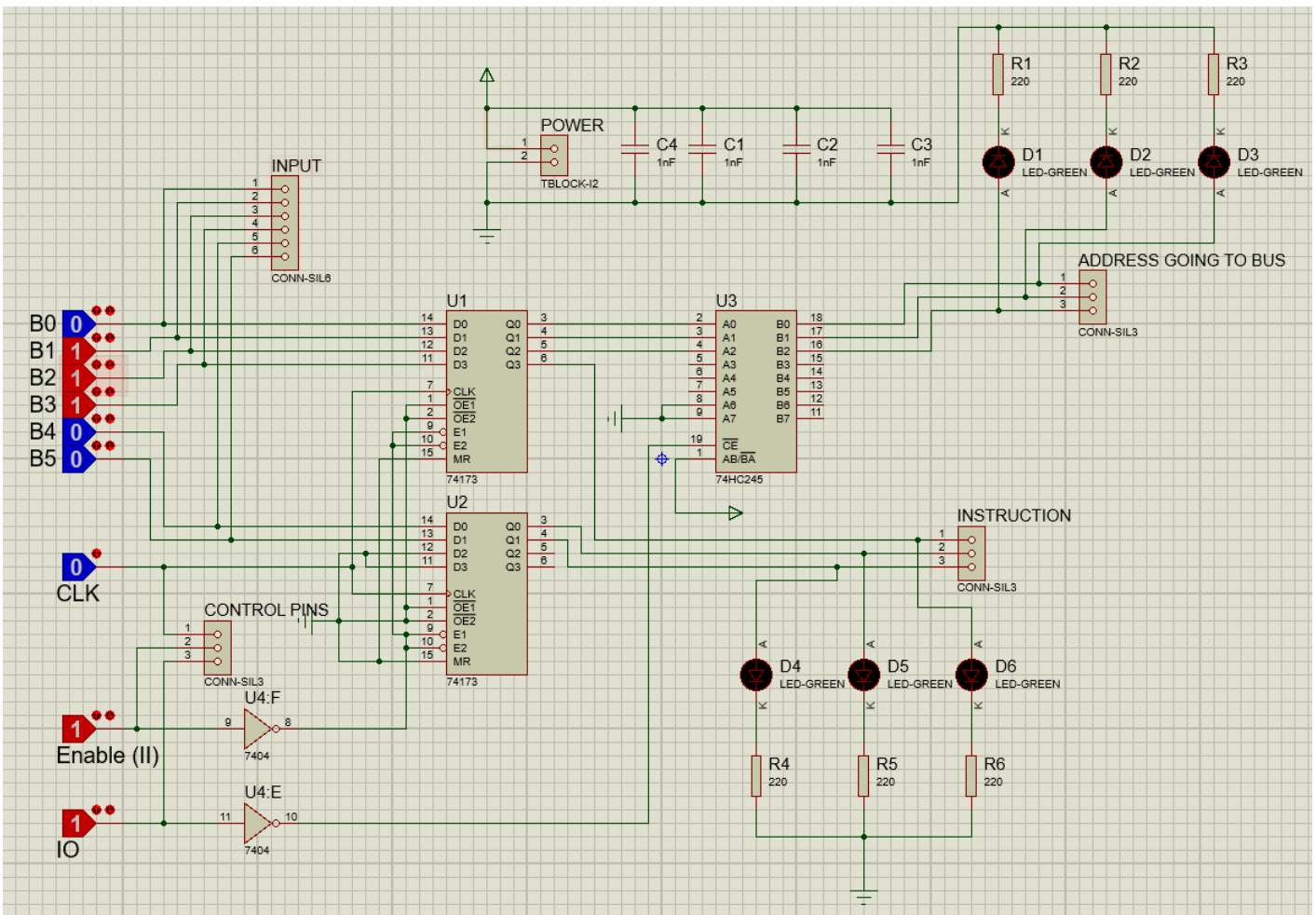
Control Bits:

- 1) **II**: Stores value from BUS
- 2) **IO**: Sends only lower nibble to BUS

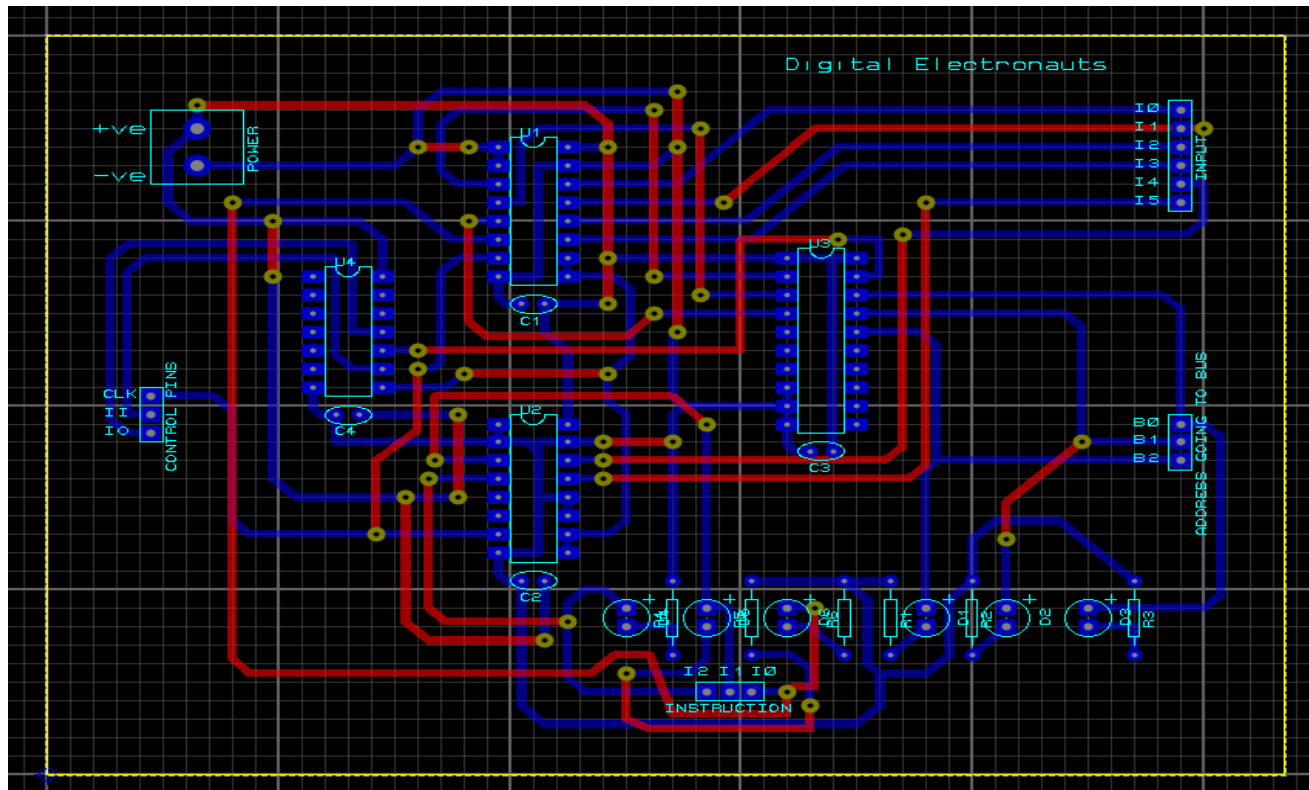
Circuit:

To build this circuit we are using two 74LS173 IC's , one 74LS245 and two 7404 (NOT Gate). 74173 is a 4-bit register and 74245 is Octal BUS transceiver (Tri-state Buffer).

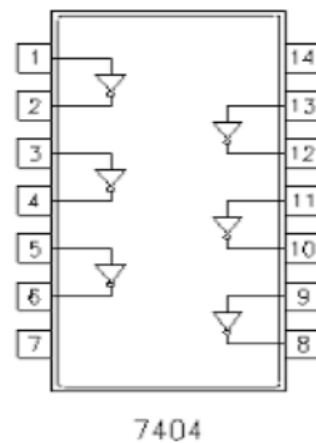
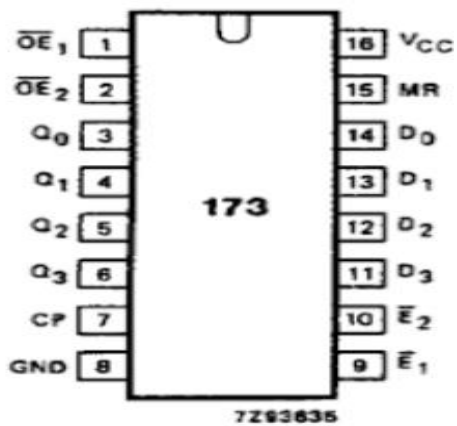
Instruction Register Circuit in Proteous:



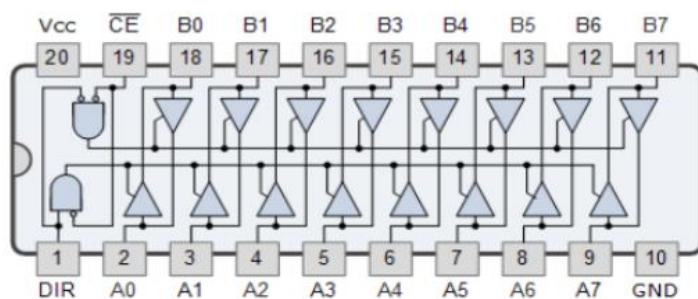
PCB :



Used IC's :



74LS245 Octal Tri-state Buffer



2) Controller Sequencer:

It is the most important module of our SAP. Without this module SAP can not operate. Every other module needs Input Enable and output Enable to operate. This enable signals have to sent in a specific sequence to all the other module. Controller sequencer does this for us. It generates 13 signals in a specific sequence according to the opcode. It synchronizes all the module so that SAP can run perfectly.

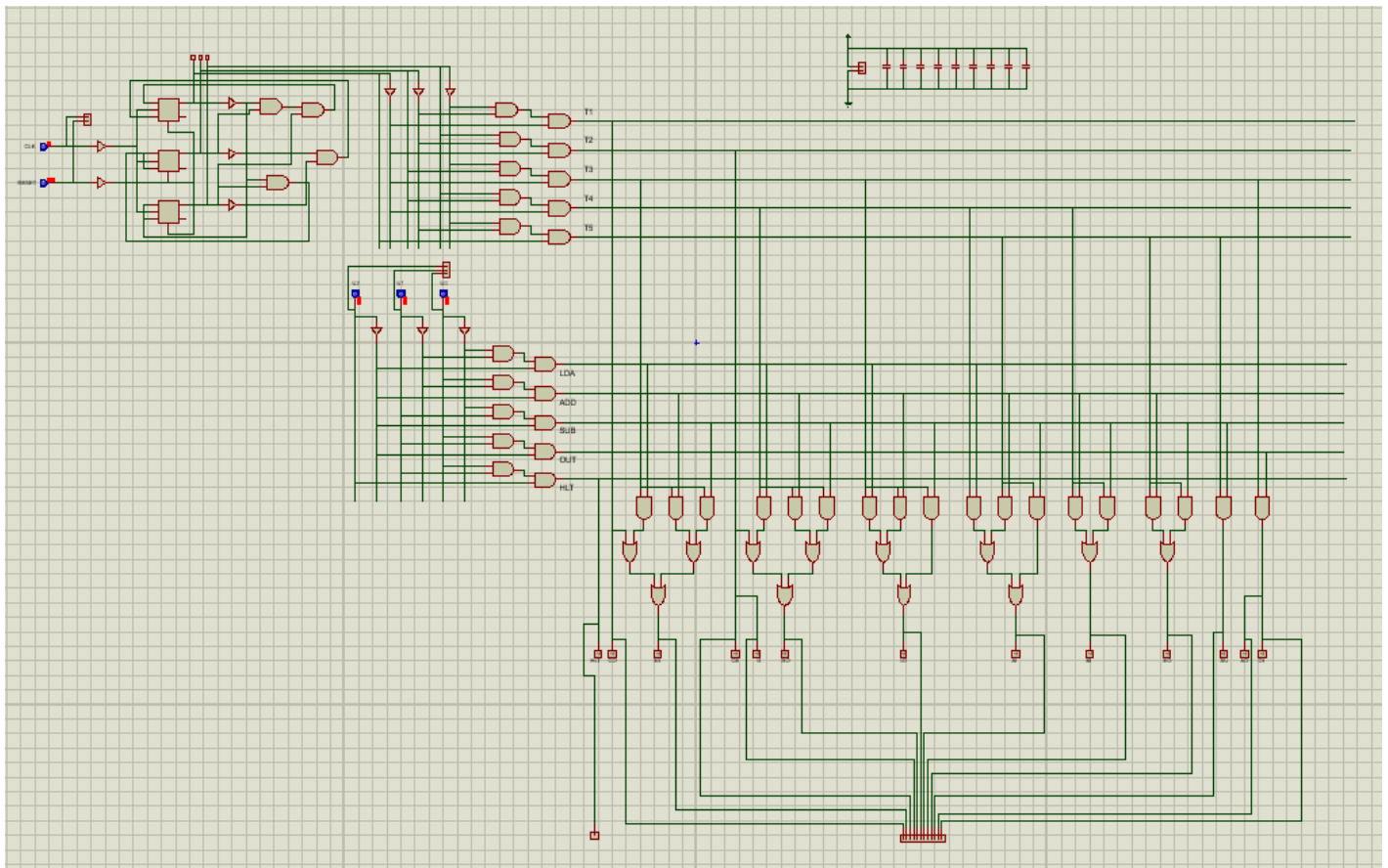
Control Bits:

- 1) **RESET:** Resets the current data

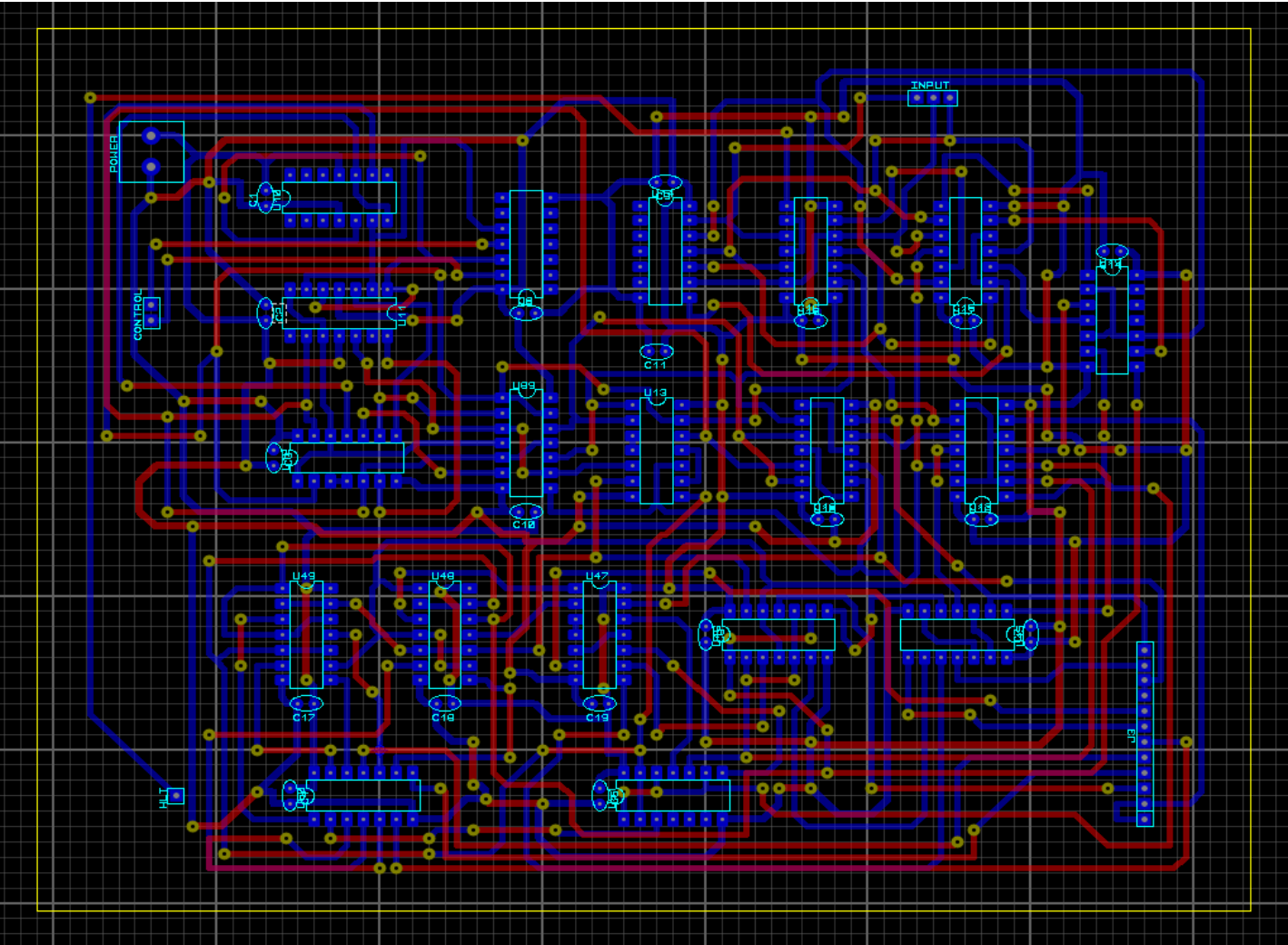
Circuit:

To built this we have used three 7473 (J-K Flip-Flop),7404, 7408,7432 IC's and capacitors.

Circuit in Proteous:



PCB:



Calculation for counter of the sequencer:

Present state	Next state	F/F inputs
$Q_2 Q_1 Q_0$	$Q_2 Q_1 Q_0$	$J_2 K_2 \quad J_1 K_1 \quad J_0 K_0$
0 0 0	0 0 1	0 X 0 X 1 X
0 0 1	0 1 0	0 X 1 X X 1
0 1 0	0 1 1	0 X X 0 1 X
0 1 1	1 0 0	1 X X 1 X 1
1 0 0	0 0 0	X 1 0 X 0 X

$J_2:$	$K_2:$	$J_1:$	$K_1:$
$Q_2 Q_1 \backslash Q_0$	$Q_2 Q_1 \backslash Q_0$	$Q_2 Q_1 \backslash Q_0$	$Q_2 Q_1 \backslash Q_0$
00 0 0	00 X X	00 0 0	00 X X
01 0 1	01 X X	01 X X	01 0 1
11 1 0	11 1 0	11 1 0	11 1 0
10 X 1	10 1 1	10 0 1	10 X 1
$J_2 = Q_2' Q_1 Q_0$	$K_2 = Q_1' Q_0'$	$J_1 = Q_2' Q_0$	$K_1 = Q_2' Q_0$

$J_0:$	$K_0:$
$Q_2 Q_1 \backslash Q_0$	$Q_2 Q_1 \backslash Q_0$
00 1 0	00 X 1
01 1 1	01 X 1
11 0 0	11 X 1
10 0 1	10 X 1
$J_0 = Q_2'$	$K_0 = Q_2$

Problem faced:

- 1) Selecting the proper IC's
- 2) Component placement
- 3) PCB design
- 4) Some components don't have PCB package
- 5) While routing the PCB

Discussion:

In this phase we designed Instruction register and controller sequencer of our SAP. Among all the module controller sequencer designing was the hardest one and instruction register was the easiest one. It is a 6-bit SAP, we can develop it further to 8 bit, as we same number of components and concepts to built 8 bit SAP. Through this phase we learnt how controller sequencer generates necessary signals for the all other module. It was a great learning phase for us.

Contribution of each member is attached with phase-E report.